

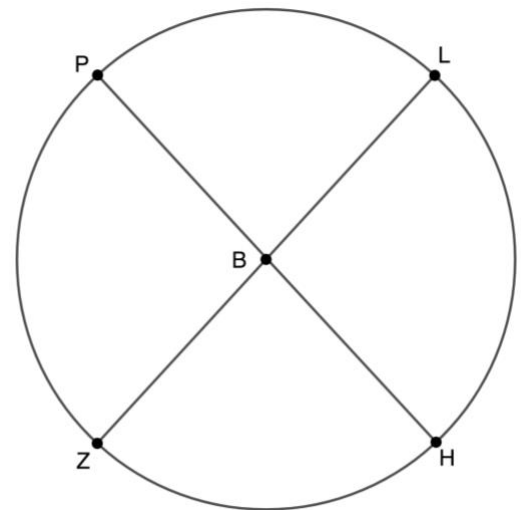
Geometry  
Unit 10  
Lesson 5 Practice

Name \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

- Circle  $C$  has points  $X, N$ , and  $P$  on the circle with  $m\angle NCX = 47.1^\circ$ . If  $m\widehat{NP} = 116.9^\circ$  and  $\widehat{NP}$  does not intersect  $\widehat{NX}$ , what is  $m\angle PCX$ ?
- Circle  $S$  has radii  $\overline{ST}$  and  $\overline{SR}$ . Find the value of  $x$  if  $m\angle RST = 135^\circ$  and  $m\widehat{RT} = (6x + 9)^\circ$ .

For questions 3-4, Circle  $B$  has diameters  $\overline{PH}$  and  $\overline{LZ}$  that both intersect at point  $B$ .

- Given  $m\widehat{HZ} = 72^\circ$ , find the  $m\widehat{PL}$ .
- Given  $m\widehat{HL} = 108^\circ$ , find the  $m\widehat{ZP}$ .

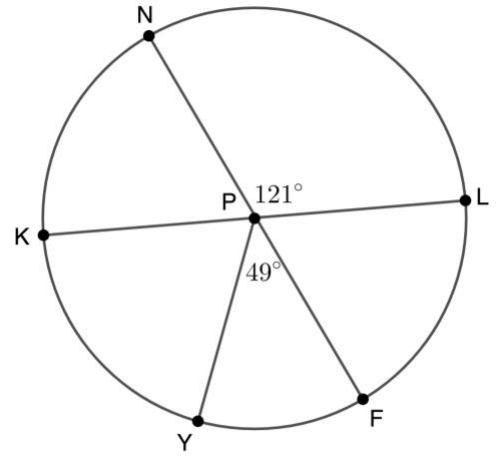


For questions 5-6, consider the following context.

Lorenzo is creating a folding fan. He needs to determine the measure of the arc of the fan before he can cut the paper. He uses the expression  $(6x + 110)^\circ$  as the measure of the central angle,  $\angle ARQ$ . For the  $m\widehat{QA}$ , he uses the expression  $(24x + 38)^\circ$ .

- What is the value of  $x$ ?
- What is the degree measure of the arc of the fan,  $\widehat{QA}$ ?

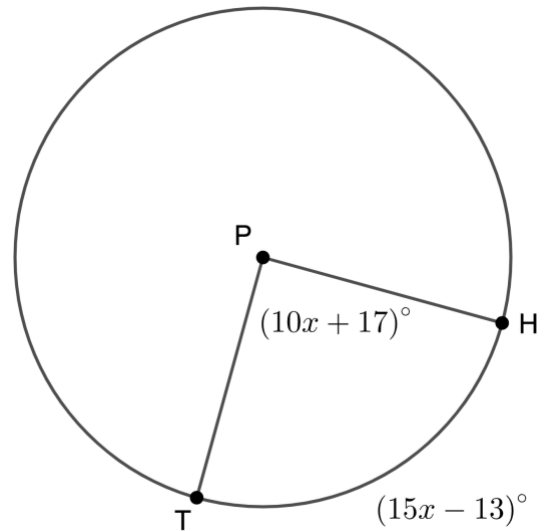
7. In Circle  $P$ ,  $m\angle NPL = 121^\circ$  and  $m\angle FPY = 49^\circ$ . Calculate the  $m\widehat{NLY}$ , given that  $P$  is the center and  $\overline{NF}$  and  $\overline{KL}$  are diameters.



8. Circle  $S$  has points  $K, D$ , and  $Y$  on the circle. The measure of the central angle,  $\angle YSD = 71^\circ$ ,  $m\widehat{KY} = 132^\circ$ , and  $\widehat{KY}$  does not intersect  $\widehat{YD}$ . Find  $m\angle KSD$ .

For questions 9-10, consider the following context.

A horse is tied to a post at point  $P$  and is grazing in a circular pattern. She decides to walk from her position at point  $H$  to the trough of water at point  $T$  along the circumference of the circular area she can roam. The measure of the central angle,  $\angle TPH$ , is  $(10x + 17)^\circ$ , and the measure of the arc that the horse walked,  $\widehat{TH}$ , is  $(15x - 13)^\circ$ .



9. Find the value of  $x$ .
10. Find the measure of the arc that the horse walked,  $\widehat{TH}$ .